Dr Syed Muzahir Abbas

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Address : 206/161 Epping Road, Macquarie Park, NSW 2113
Working Rights : Full-Time (Australian Citizen)

Academic Qualifications

- **Ph.D. Electronics Engineering** ***(2012-2015)***
  Macquarie University, Sydney, NSW, Australia.
  Scholarships: iMQRES and CSIRO Postgraduate Scholarships
  Thesis: Antennas and Components for Modern Telecommunication Systems including Wireless Body Area Networks
- **MSc. Computer Engineering** ***(2008-2009)***
  CASE (Center for Advanced Studies in Engineering), Islamabad, Pakistan.
- **BSc. Electrical Engineering (Telecommunication)** ***(2002-2006)***
  COMSATS Institute of Information Technology, Islamabad, Pakistan.

Skill Set

- **Hardware:** Network Analyzer, Anechoic Chamber, PCB Milling, Microwave Links, SDH-E1-Fiber Multiplexers, Power Meter, E1 tester, Power Supply, Oscilloscope, Function Generator, Reconfigurable Electronics,
- **Software:** CST Microwave Studio, ANSYS HFSS, CREO, AWR, Draft Sight, Simulink, MATLAB, C++, Java, Python, Pspice, Verilog HDL, LATEX, Sigmaplot, Microsoft office.

Employment & Experience

- **Lead Antenna Design Engineer** **(Jan 2023 to Date)**
  GME, NSW, Australia
  - Research and development, antennas for wireless mobility solutions, including UHF, VHF, 4G and 5G
  - Product development, testing and analysis, troubleshooting, root cause analysis, automation consideration, support to Production
  - Project management, risk evaluation, preparation of technical reports and documentation
- **Adjunct Fellow/Research Supervisor (Engineering)** **(2016 to Date)**
  Macquarie University, NSW, Australia
  - Research supervision of PhD and Masters degree program scholars
  - Research and development
  - Publications and technical write-ups
  - Funding grants/application and nation/international collaborations
- **Lecturer (Engineering/Telecommunication)** **(2019 to Date)**
  University of Sydney, NSW, Australia
  - Lecturing course module of Masters degree programs for Elec 5101: Antennas and Propagation
  - Conduction of laboratories/tutorials
  - Capston Projects/Research suervision
- **Research Engineer** **(Nov 2022 – Jun 2023)**
  Universiy of Technology Sydney, NSW, Australia
  - Research and development
  - Project management, preparation of technical reports and documentation
- **Senior Principal Engineer (RF/Electronics)** **(Jun 2020 – Jan 2023)**
  BENELEC Technologies, NSW, Australia
  - Heading R & D for wireless mobility solutions, including UHF, VHF, 4G and 5G
  - Responsible for production, qualility control and quality assurance

- Testing and analysis, troubleshooting, root cause analysis, automation consideration, supervision, support to New Product Intoduction (NPI) process
- Project management, team management, RF Planning, risk evaluation, preparation of technical reports and documentation

- **Senior Antenna Design Engineer** **(Nov 2019 – Jun 2020)**
  BENELEC Technologies, NSW, Australia
  - Research, design and development of wireless mobility solutions, including UHF, VHF, 4G and 5G
  - Product development for Austrailan and global clients
  - Testing and analysis, troubleshooting, root cause analysis, automation consideration, supervision, support to New Product Intoduction (NPI) process
  - RF Planning, project management, risk evaluation, preparation of technical reports and documentation
- **Lecturer (Engineering)** **(Feb 2018 – Nov 2019)**
  Western Sydney University, University of Sydney,
  - Lecturing course module of Bachelors degree programs for courses, Electronics, Electromagnetics Engineering, Communication Systems
  - Conduction of laboratories and tutorials
  - Course coordination and Lead Lecturer
- **Lecturer (Information Technology)** **(Feb 2018 – Nov 2019)**
  (Charles Sturt University, King's Own Institute, ATMC-Federation University, ATMC-University of Sunshine Coast), NSW, Australia
  - Lecturing course module of Bachelors and Masters degree programs for Information and Communications Technology, Research Ethics and Professional Practices
  - Administrative tasks, counseling, conduction of laboratories and tutorials
  - Subject coordination
- **RF Engineer II (R & D, Mobility Solutions)** **(Jul 2016 – Feb 2018)**
  CommScope, NSW, Australia
  - Research, design and development of wireless mobility solutions, including base station antennas
  - Product development including First in the Family Products on global level
  - Testing and analysis, troubleshooting, root cause analysis, automation consideration, supervision, support to New Product Intoduction (NPI) process
  - Project management, risk evaluation, preparation of technical reports and documentation
- **RF Consultant** **(Jun 2015 – Jul 2016)**
  Future Technologies & Innovations, NSW, Australia
  - Carried out RF consultancy work of various clients including NSW Telco Authority, NSW, Australia
  - Project management and interaction with clients
  - RF planning using Mentum Planet
  - Coverage optimization
- **Researcher/Head Teaching Assistant** **(Jul 2012 – Jul 2016)**
  Macquarie University, NSW, Australia
  - Courses: Introduction to Electronic Systems, Linear Circuits and Devices, Introduction to Mechanical Engineering, Introduction to Engineering, Electromechanics, Engineering Practice
  - Promoted to Head TA in Sep' 14: Coordinated with course conveners and supervised five other TAs.
- **IELTS Invigilator/Marker** **(Jan 2013 *to date*)**
  Access Macquarie, NSW, Australia
  - Invigilation and administrative support / Chief Invigilator
  - Marking of IELTS tests
- **Lecturer (Electrical Engineering)** **(Oct 2008 – Feb 2012)**
  COMSATS Institute of Information Technology, Islamabad, Pakistan
  - Supervision of teaching staff and thesis/project supervision of postgraduate/undergraduate students
  - Teaching, research, conduction and management of engineering laboratories

- Administrative duties (degree accreditation, thesis examinations and management, teaching quality assurance, upgradation of laboratories)

- **Transmission Engineer** **(Jul 2007 – Oct 2008)**
  ALCATEL-LUCENT, Islamabad, Pakistan
  - Installation, configuration, commissioning, operation and maintenance of SDH/PDH Links, repeater sites, multiplexer (E1s to Optical Fiber/Transport Network), test equipments
  - Site inspection, follow-ups, coordination and interaction with customers
  - IP planning, installation, DCN connectivity of microwave links for Network Management System (NMS)
  - Technical Site Survey Reports for Transmission, line of sight surveys for microwave links and coverage

Memberships

- Honorary Research Fellow, Macquarie University, NSW, Australia
- IEEE (Institute of Electrical and Electronics Engineers)
- IEEE Antenna and Propagation Society (AP-S)
- IEEE Microwave Theory and Techniques Society (MTT-S)
- IEEE Electromagnetics Competibility Society (EMC-S)
- IEEE Young Professionals
- Australian Association for Engineering Education (AAEE), Australia

Volunteer Positions Held

- 2026: Chair, IEEE NSW Section
- 2026: Vice Chair, IEEE MTT-S Young Professionals
- 2026: Chair, IEEE MTT-S Young Professionals Outstanding Lecturers Program
- 2026: Chair, Educational Resourse Development IEEE MTT-S Young Professionals
- 2024-25: Chair, IEEE NSW Section
- 2024-25: R-10 Coordinator, IEEE MTT-S Young Professionals
- 2024-25: ANZ Coordinator, IEEE Young Professionals (Region 10)
- 2023: Vice Chair, IEEE NSW Section
- 2020-23: Chair, IEEE NSW AP/MTT-S Chapter
- 2020-23: ANZ Coordinator, IEEE Young Professionals (Region 10)
- 2020-25: Webmaster, IEEE NSW Section
- 2020: Secretary, IEEE NSW Young Professionals
- 2019: Chair, IEEE NSW Young Professionals
- 2019: Vice Chair, IEEE NSW AP/MTT-S Chapter
- 2017 - 2019: Webmaster, IEEE NSW Section
- 2017 - 2019: Committee Member, IEEE NSW Section
- 2018: Chair, IEEE NSW Young Professionals
- 2018: Student Activities Chair, IEEE Australia Council
- 2018: Vice Chair, IEEE NSW AP/MTT-S Chapter
- 2018: Program Chair/Webmaster, IEEE TENSYMP 4-6 July 2018, Sydney
- 2017: Vice Chair/Webmaster, IEEE Australia New Zealand Students and Young Professionals Congress (ANZSCON) 2017
- 2017: Vice Chair, IEEE NSW Young Professionals
- 2017: Treasurer, IEEE NSW AP/MTT-S Chapter
- 2016: Secretary, IEEE NSW Young Professionals
- 2016: Committee Member, IEEE NSW AP/MTT-S Chapter
- 2013-2015: Chair, IEEE MTT-S Student Chapter at Macquarie University
- 2013-2014: Chair, IEEE Student Branch at Macquarie University. Won First Position in 2013 IEEE Australia Council Outstanding Student Branch and Won Second Position in 2014 IEEE Australia Council Outstanding Student Branch.
- 2012-2015: Editorial Board, IEEE Student Branch at Macquarie University
- 2013-2014: Postgraduate representative from Faculty of Science on Macquarie University Student Advisory Board

REVIEWER (International Research Journals)

- IEEE Transactions on Antennas and Propagation
- IET Microwaves, Antennas & Propagation Journal
- IEEE Antennas and Propagation Magazine
- IEEE Antennas and Wireless Propagation Letters
- Microwave and Optical Technology Letters

- IEEE Transactions on Terahertz Science and Technology
- IEEE Access
- International Journal of Microwave and Wireless Technologies
- IEEE IoT Journal
- IETE Journal of Research
- Journal of Electromagnetic Waves and Applications
- IEEE Electronics Letters
- MDPI Journal of Composites Science
- MDPI Electronics
- MDPI Energies
- MDPI Textiles
- MDPI Applied Sciences
- MDPI Sensors
- MDPI Diagnostics
- MDPI Signals
- MDPI Biosensors
- MDPI Micromachines
- MDPI Crystals
- MDPI Remotesensing
- MDPI Diagnostics
- IEEE Internet of Things Journal
- IEEE Journal of Electromagnetics, RF, and Microwaves in Medicine and Biology
- Progress in Electromagnetic Research
- Flexible and Printed Electronics
- Wireless Personal Communications (Springer)
- Cellulose International Journal (Springer)
- Journal of Electromagnetic Engineering and Science
- Optics and Laser Technology

Conferences (Committee Member)

- 2027: Conference Executive Chair, International Conference on Artificial Intelligence, Computer, Data Sciences and Applications (ACDSA 2027), Rio de Janeiro, Brazil,, 2-4 April, 2027.
- 2026: Young Professional and Student Events Chair, IEEE MTT-S International Microwave Workshop Series on Advanced Materials and Processes for RF and THz Applications (IMWS-AMP 2026), Cairns, Australia, 22-24 July, 2026.
- 2026: Publication Chair, International Conference on Electrical, Computer, Communications and Mechatronics Engineering (ICECCME 2026), Bali, Indonesia, 15-16 October, 2026.
- 2026: Publication Chair, International Conference on Electrical, Computer and Energy Technologies (ICECET 2026), Rome, Italy, 6-8 July, 2026.
- 2026: Conference Executive Chair, International Conference on Artificial Intelligence, Computer, Data Sciences and Applications (ACDSA 2026), Boracay Island, Philippines, 2-4 April, 2026.
- 2025: Co-Chair, IEEE International Conference on Future Machine Learning and Data Science (FMLDS), Sydney, Australia, 20-23 November, 2024.
- 2025: Executive Chair, International Conference on Electrical and Computer Engineering Researches (ICECER 2025), Madagascar, 6-8 December, 2025.
- 2025: Local Organizing Committee Member, URSI Asia-Pacific Radio Science Conference (URSI AP-RASC 2025), Sydney, Australia, 17-22 August, 2025.
- 2025: Publication Chair, International Conference on Artificial Intelligence, Computer, Data Sciences and Applications (ACDSA 2025), Antalya, Turkiye, 7-9 August, 2025.
- 2025: International Advisory Committee, International Conference on Microwave, Antennas & Circuits (ICMAC 2025), Islamabad, Pakistan, 16-17 April, 2025.
- 2025: Technical Program Committee Member, International Conference on Emerging Systems and Advanced Technology (ICoESAT 2025), Saudi Arabia, 14-15 April, 2025.
- 2024: General Chair, International Conference on Electrical, Computer and Energy Technologies (ICECET 2024), Sydney, Australia, 25-27 July, 2024.

- 2024: International Advisory Committee, International Conference on Communication Engineering and Emerging Technologies (ICoCET 2024), Penang, Malaysia, 2-3 September 2024.
- 2024: Technical Program Co-Chair, International Conference on Sensing Technology (ICST 2024), Sydney, Australia, 9-11 December, 2024.
- 2023: Publication Chair, International Conference on Electrical, Computer and Energy Technologies (ICECET 2023), 16-17 November, Cape Town, South Africa, 16-17 November, 2023.
- 2022: Local Arrangement Chair, 27th International Symposium on Antennas and Propagation (ISAP 2022), Sydney, Australia, 31 October - 3 November, 2022.
- 2021: Special Session organizer on 5G, European Conference on Antennas and Propagation (EuCAP 2021), Düsseldorf, Germany, 22-26 March, 2021.
- 2020: Judge of the Best Paper Committee: Asia-Pacific Microwave Conference (APMC 2020), Hong Kong, China, 8-11 December, 2020.
- 2020: Technical Programme Committee: 14th International Conference on Sensing Technology (ICST 2020), Nagano, Japan, 25-27 November, 2020.
- 2020: Technical Programme Committee: The Asia-Pacific Symposium on Applied Electromagnetics and Mechanics (APSAEM 2020), Melaka, Malaysia, 14-17 July, 2020.
- 2020: Meta Reviewer for European Conference on Antennas and Propagation (EuCAP 2020), 15-20 March, 2020 in Copenhagen, Denmark.
- 2019: Technical Programme Committee, International Conference on Electrical Engineering Research and Practice (ICEERP 2019), Sydney, Australia, 24-28 November, 2019, also helped Western Sydney University to get IEEE NSW section sponsorship.
- 2019: Technical Programme Committee: 13th International Conference on Sensing Technology (ICST 2019), Sydney, Australia, 2-4 December, 2019.
- 2018: Represented IEEE NSW section as Young Professional on Committee for IEEE Raising Stars Conference, USA. Attended their weekly meeting and contributed to various planning activities.
- 2018: Program Chair/Webmaster, IEEE TENSYMP 4-6 July 2018, Sydney, Australia
- 2017: Vice Chair/Webmaster, IEEE Australia New Zealand Students and Young Professionals Congress (ANZSCON 2017), Sydney, Australia

Personal Honours

- 2025: IEEE EMC-S Young Professionals Ambassador
- 2023: Best Poster Award, IEEE International Symposium on Antennas and Propagation (ISAP 2023), Malaysia
- 2023: Best (2nd) IEEE AP-S Chapter Award, for IEEE NSW AP/MTT Joint Chapter as Chair
- 2022: Best IEEE MTT-S Chapter Award, for IEEE NSW AP/MTT Joint Chapter as Chair
- 2020: Best Paper Award, 2020 IEEE 5G World Forum (5GWF'20)
- 2019: IEEE NSW Young Professional Award of the Year 2019
- 2018: Young Scientist Award 2018 by the International Union of Radio Science - Atlantic Radio Science Conference (URSI AT-RASC)
- 2013-2015: CSIRO Postgraduate Fellowship, CSIRO, Australia
- 2012-2015: International Macquarie University Research Excellence Scholarship (iMQRES) for PhD, Macquarie University, Australia
- 2015: Visiting Researcher at Queen Mary, University of London, London, UK
- 2014: Visiting Researcher at Electro Science Laboratory, The Ohio State University, Columbus, Ohio, USA
- 2012: Research Productivity Award, COMSATS Institute of Information Technology, Islamabad, Pakistan
- 2010: Research Productivity Award, COMSATS Institute of Information Technology, Islamabad, Pakistan

Grants Received

- CSIRO Postgraduate Studentship Scholarship, 2013-2015 **(AUD $30,000)**: CSIRO, Australia.
- International Macquarie University Research Excellence Scholarship (iMQRES) for PhD, 2012-2015 **(AUD $208,108)**: Macquarie University, Australia.
- 2 WiMed Projects Grants 2015 **(AUD $5000)**: Macquarie University, Australia.

- WiMed Travel Grant 2015 **(AUD $1000)**: Macquarie University, Australia.
- MQ PGRF 2015 for Europe **(AUD $5000)**: Macquarie University, Australia.
- FSE PGRF 2014 for North America **(AUD $5000)**: Macquarie University, Australia.
- WiMed Projects Grant 2014 **(AUD $5000)**: Macquarie University, Australia.

Patent Applications:

1. Base Station Antennas having Parasitic Coupling Units - (US Application No. 62/505174)
2. Electromagnetically Coupled Printed Antenna for Sub-6GHz 5G Applications (AU Patent No. 2021107352)
3. 5G Shark Fin Style Antenna for Vehicular Applications (AU Patent No. 2021107356)
4. 4 Port MIMO Antenna with Orthogonal Arrangement for WiFi and 5G Communications (AU Patent No. 2021107360)
5. 6 Port MIMO Antenna for GPS/GNSS, WiFi and 5G Communication Systems (AU Patent No. 2021107367)
6. 8 Port MIMO Antenna for GPS/GNSS, WiFi and 5G Communication Systems (AU Patent No. 2021107500)
7. Compact Terminal Antenna for 5G Applications – (AU Patent No. 2020101214)
8. Dual Band Ground Dependent Flexible Antenna for VHF/UHF Applications (AU Patent No. 2020100788)
9. VHF High Band Directional Antenna with Broadband Characteristics for In-building Radio Systems using Bidirectional Repeater Technologies (AU Patent No. 2021102942)
10. Colinear Base Station Antenna for Point to Multipoint Linking and Public Mobile Radio Applications (AU Patent No. 2020103834)
11. Omni-directional VHF Base Station Antenna for LMR Communications (AU Patent No. 2021101089)
12. Vertical Broadband UHF Base Station Antenna with Stable Radiation Patterns (AU Patent No. 2021100859)
13. UHF Broadband Marine Antenna with Heavy Duty Radome and Short Mounting Tube for Harsh Environmental Conditions (AU Patent No. 2020103688)
14. Stacked Folded Dipole Array with Orthogonal Arrangement having Omni-directional Characteristics for UHF Applications (AU Patent No. 2020101828)
15. 220MHz VHF Broadband Omni-directional Base Station Antenna (AU Patent No. 2021105243)
16. UHF Broadband Antenna for In-building Coverage using Distributive Antenna System (AU Patent No. 2021105242)
17. Broadband VHF Antenna with Connector Secured at the End of Boom for Point-to-Point Link Applications (AU Patent No. 2021106988)
18. Passive Intermodulation (PIM) Test Bed (AU Patent No. 2021106985)
19. Smart Control Configuration for Power Supply with Safety Features and Real-time Monitoring (AU Patent No. 2021107415)
20. Cut off Configuration using FET and Driver IC for Enhanced Battery Efficiency (AU Patent No. 2021107403)

Publications:

Book/Book Chapters

1. **Syed Muzahir Abbas**, and Yang Yang, "Recent Advances in Antennas and Millimeter-Wave Applications for Mobile Communication Systems", edited by Syed Muzahir Abbas, and Yang Yang, MDPI Electronics, 2024, https://doi.org/10.3390/books978-3-7258-0839-7 ***(Edited Book)***
2. M. A. B. Abbasi, M. Rizwan, S. Shahid, S. Rafique, H. T. Awan, and **S. M. Abbas**, "A Compact Multiband Antenna for GSM and WiMAX Applications Emerging Trends and Applications in Information Communication Technologies", vol. 281, B. S. Chowdhry, F. K. Shaikh, D. M. A. Hussain, and M. A. Uqaili, Eds., ed: Springer Berlin Heidelberg, 2012, pp. 20-30. ***(Book Chapter)***
3. **S. M. Abbas**, I. Saleem, A. Parveen, H. Zahra, and S. A. Khan, "Chebyshev Multiband Patch Antenna Emerging Trends and Applications in Information Communication Technologies", vol. 281, B. S. Chowdhry, F. K. Shaikh, D. M. A. Hussain, and M. A. Uqaili, Eds., ed: Springer Berlin Heidelberg, 2012, pp. 10-19. ***(Book Chapter)***

International Journal Publications

1. M. Hussain, W. A. Awan, **S. M. Abbas**, and Y. Zhu, “Compact Bendable Multiwalled Carbon Nanotube/PDMS Antenna for WLAN and Wi-Fi 6E Wearables in 5G/6G Networks,” Advanced Engineering Materials, vol. 27, no. 23, 2025, https://doi.org/10.1002/adem.202502127
2. H. Anam, **S. M. Abbas**, I. B. Collings and S. C. Mukhopadhyay, "Development of RFID Humidity Sensor and Radar Cross Section Stability," in *IEEE Sensors Journal*, vol. 25, no. 13, pp. 25595-25604, 2025, doi: 10.1109/JSEN.2025.3573268.
3. **S. M. Abbas**, A. P. Aji, K. Wu and J. -C. Chiao, "Two Special MTT-S Inter-Society Technology Panels at the 2024 Asia-Pacific Microwave Conference [MTT-S Society News]," in *IEEE Microwave Magazine*, vol. 26, no. 7, pp. 21-23, July 2025, doi: 10.1109/MMM.2025.3554952.
4. M. Hussain, W. A. Awan, H. Zahra, **S. M. Abbas**, Samir Salem Al-Bawri, and Yong Zhu,”Design and characterization of multiwall carbon nanotube/polydimethylsiloxane composite MIMO antenna for wearable and vehicular applications,” *Scientific Reports,* vol. 15, no. 1, 2025. https://doi.org/10.1038/s41598-025-13112-1
5. H. Anam, **S. M. Abbas**, I. B. Collings, and S. Mukhopadhyay, “A PDMS/MWCNTs RFID flexible tag with advanced resonator design for read range enhancement in IoT monitoring systems,” *Scientific Reports,* vol. 15, no. 1, 2025. https://doi.org/10.1038/s41598-025-86773-7
6. H. Anam, **S. M. Abbas**, I. B. Collings, and S. Mukhopadhyay, “Materials-Driven Advancements in Chipless Radio-Frequency Identification and Antenna Technologies,” *Sensors (Basel)*, vol. 25, no. 9, 2025. doi: 10.3390/s25092867.
7. Y. Zhu, S. Jalalifar, H. P. Phan, **S. M. Abbas** and M. Asadnia, "A Smart Multisensor Drowning Detection Device With Real-Time Alarm Function," in *IEEE Sensors Journal*, vol. 25, no. 4, 2025. doi: 10.1109/JSEN.2024.3518436.
8. A. Parveen, M. M. Makhdoom, K. Ismail, G. Murtaza, H. Ali, K. H. Shah, N. S. Abd El-Gawaad , and **S. M. Abbas**, “Ab Initio Calculation of Half-Metallicity, Magnetic and Transport Properties of Potassium-Based $KLAO_2$ (LA = Ho^{3+}, Tm^{3+}, Yb^{3+}) Oxides for Spintronics Devices,” *Journal of Electronic Materials,* 2025. https://doi.org/10.1007/s11664-025-11919-2.
9. M. Hussain, W. A. Awan, **S. M. Abbas**, and Y. Zhu, “Design and Development of Low-Profile Polymer Based Broadband Antenna for On-Body Applications,” *Results in Engineering*, vol. 25, 2025. https://doi.org/10.1016/j.rineng.2024.103818.
10. N. Nasir, M. H. Jamaluddin, N. H. Shahadan, **S. M. Abbas**, H. Ahmad, “Compact Substrate Integrated Waveguide Cubical Dielectric Resonator Antenna for fifth-generation applications,” *AEU - International Journal of Electronics and Communications*, vol. 187, 2024. https://doi.org/10.1016/j.aeue.2024.155551.
11. M. A. S. Syed, S. Mohsin, J. Yu, Y. Yao, M. A. B. Abbasi and **S. M. Abbas**, "Comparative Analysis of Novel Gliding Walled Multilayer and Traditional Longitudinally Corrugated Waveguides for W-Band Millimeter-Wave Applications," in *IEEE Access*, vol. 12, pp. 138827-138834, 2024. doi: 10.1109/ACCESS.2024.3437413
12. M. Ayaz, I. Ullah, A. Iftikhar, **S. M. Abbas**, B. D. Braaten, S. Khattak, and M. Alathbah, "A Flexible Conformal Phased Array With Embedded Magnetic Particles Based Composite Right/Left Handed Metamaterial Phase Shifters," in *IEEE Access*, vol. 12, pp. 129147-129159, 2024. doi: 10.1109/ACCESS.2024.3452185.
13. B. Hayat, J. Zhang, A. Khan, **S. M, Abbas**, A. Majeed, and S. S. Al-Bawri, "A Flexible and Optical Transparent Metasurface Absorber with Broadband RCS Reduction Characteristics," *Nanomaterials* 14, no. 18: 1507, 2024. https://doi.org/10.3390/nano14181507
14. M. Hussain, H. Zahra, **S. M. Abbas**, and Y. Zhu, “Flexible Dielectric Materials: Potential and Applications in Antennas and RF Sensors”, Advanced Electronic Materials, 2400240, 2024. https://doi.org/10.1002/aelm.202400240
15. H. A. Khan, A. Majeed, H. Zahra, F. G. Kakepoto, **S. M. Abbas**, and M. Alathbah, “Transparent conformal metasurface absorber for ultrawideband radar cross section reduction”, Journal of Physics D: Applied Physics, 57 135105, 2024. doi: 10.1088/1361-6463/ad1951
16. M. Nasir, A. Iftikhar, **S. M. Abbas**, R. Saleem, M. F. Shafique and M. Alathbah, "A Wideband Broadside Coupled Yagi Antenna and Arrays System for Ku Band Applications," in IEEE Access, vol. 11, pp. 126967-126978, 2023, doi: 10.1109/ACCESS.2023.3331388.
17. K. Ismail, A. Parveen, A. Zia, S. Razzaq, G. Murtaz, **S. M. Abbas**, and J.Foroughi, “Ab initio calculations of structural, electronic, optical, and magnetic properties of delafossite SMoO2 (S = Na, K, Rb, Cs) for spintronics”. Applied Physics A, 129, 676, 2023. https://doi.org/10.1007/s00339-023-06944-z
18. I. Saleem, U. Rafique, S. Agarwal, H. S. Savci, **S. M. Abbas**, and S. Mukhopadhyay, “Ultra-Wideband Fractal Ring Antenna for Biomedical Applications”, International Journal of Antennas and Propagation, 2023, doi: 10.1155/2023/5515263
19. M. U. Tahir, U. Rafique, M. M. Ahmed, **S. M. Abbas**, S. Iqbal and S. W. Wong, “High gain metasurface integrated millimeter-wave planar antenna,” International Journal of Microwave and Wireless Technologies, pp. 1–12, 2023. doi:10.1017/S1759078723000934
20. V. Rastinasab, W. Hu, W. Shahzad, and S**. M. Abbas,** “CubeSat-Based Observations of Lunar Ice Water Using a 183 GHz Horn Antenna: Design and Optimization,” *Applied Sciences*, vol. 13, no. 16, p. 9364, Aug. 2023, doi: 10.3390/app13169364.

21. S. Shahid, H. Zahra, S. B. Qaisar, I. H. Naqvi, **S. M. Abbas**, and S. Mukhopadhyay, “Radio Link Model for Node Deployment in Underground Mine Sensor Networks,” *Applied Sciences*, vol. 13, no. 15, p. 8987, Aug. 2023, doi: 10.3390/app13158987.

22. A. Kiyani, M. Asadnia, **S. M. Abbas**, K. P. Esselle, and A. Mahmoud, “Wide Dual-Band Circularly Polarized Diecletric Resonator: Innovative Integration of a Single Hybrid Feed and Thin Grounded Metasurface,” *Micromachines*, vol. 14, no. 7, p. 1432, Jul. 2023, doi: 10.3390/mi14071432.

23. W. Khan, S. Shahid, W. Iqbal, A. S. Rana, H. Zahra, M. Alathbah, and **S. M. Abbas**, “Semi-Coprime Array with Staggered Beam-Steering of Sub-Arrays,” *Sensors*, vol. 23, no. 12, p. 5484, Jun. 2023, doi: 10.3390/s23125484.

24. Q. Ali, Y. Xiao, S. Shafiq, W. Tan, W. Shahzad W, and **S. M. Abbas**, “1-Bit Hexagonal Meander-Shaped Wideband Electronically Reconfigurable Transmitarray for Satellite Communications,” *Electronics*, vol. 12, no. 9, p. 1957, Apr. 2023, doi: 10.3390/electronics12091957.

25. S. Shrestha, **S. M. Abbas**, M. Asadnia, and K. P. Esselle, “Enhancement of Feed Source through Three Dimensional Printing,” Micromachines, vol. 14, no. 6, p. 1244, Jun. 2023.

26. H. A. Khan, U. Rafique, **S. M. Abbas**, F. Ahmed, Y.Huang, J. A. Uqaili, and A.Mahmoud, “Polarization-Independent Ultra Wideband RCS Reduction Conformal Coding Metasurface Based on Integrated Polarization Conversion-Diffusion-Absorption Mechanism,” *Photonics*, vol. 10, no. 3, p. 281, Mar. 2023.

27. A. Yasin, N. Gogosh, S. I. Sohail, **S. M. Abbas**, M. F. Shafique, and A. Mahmoud, “Relative Permittivity Measurement of Microliter Volume Liquid Samples through Microwave Filters,” *Sensors*, vol. 23, no. 6, p. 2884, Mar. 2023.

28. H. A. Khan, C. Huang, Q. Xiao, and **S. M. Abbas**, “Polarization-Dependent Coding Metasurface with Switchable Transmission and RCS Reduction Bands,” Micromachines, vol. 14, no. 1, p. 78, 2023.

29. S. Shrestha, **S. M. Abbas**, M. Asadnia and K. P. Esselle, "Realization of Three Dimensional Printed Multi Layer Wide Band Prototype," in IEEE Access, vol. 10, pp. 130944-130954, 2022.

30. A. M. Khan, M. M. Ahmed, U. Rafique, A. Kiyani, and **S. M. Abbas**, “An Efficient Slotted Waveguide Antenna System Integrated with Inside-Grooves and Modified Gaussian Slot Distribution,” Electronics, vol. 11, no. 18, p. 2948, Sep. 2022.

31. A. A. Ibrahim, H. Zahra, O. M. Dardeer, N. Hussain, S. M. Abbas, and M. A. Abdelghany, “Slotted Antenna Array with Enhanced Radiation Characteristics for 5G 28 GHz Communications,” Electronics, vol. 11, no. 17, p. 2664, Aug. 2022.

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Thesis/ Capstone Project Supervision/Co-supervision

Postdocs : 1 PhD: 8 Postgraduate : 20 Undergraduate : 55

References (available if required)